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AIPI 590: Alternative Data

Fall 2026

Dr. Pramod Singh

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Attendance and Feedback from Google Office Visit

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Where we are

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Week	Date	Topic
Week 1	Aug 26, 2026	Foundations of Alternative Data
Week 2	Sep 2, 2026	Types of Alternative Data
Week 3	Sep 9, 2026	The AI Lifecycle in Alternative Data Products
Week 4	Sep 16, 2026	Presentation of Ideas for the Term Project
Week 5	Sep 23, 2026	Investment Research Introduction
Week 6	Sep 30, 2026	Investment Research with Alternative Data – Methods & Pitfalls
Week 7	Oct 7, 2026	Investment Research with Alternative Data – From Models to Strategy
Week 8	Oct 14, 2026	Mid-Semester Presentations
Week 9	Oct 21, 2026	Ethics and Explainability of Alternative Data Products
Week 10	Oct 28, 2026	Case Study Debate & Alternative Data Simulation
Week 11	Nov 4, 2026	Synthetic Data in Alternative Data - Expanding the Edge
Week 12	Nov 11, 2026	Industry Project Presentation
Week 13	Nov 18, 2026	Term Project Presentations

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Please welcome our guest speaker

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Meet our Speaker : Arun Jain: Ground Truth for AI



About the Speaker

- Founder and Managing Principal, Kubera Capital; author of *The Portfolio Architect* and *The Intelligent Chaos*
- Two decades in enterprise tech at Amazon, Goldman Sachs, Morgan Stanley, Credit Suisse
- Board member, TiE Raleigh Durham and ASEI; IIT engineer, UNC MBA

About the Talk

- **Problem:** AI needs a reliable link between data and reality, not just more data
- **Evidence:** Seven failures and lessons, from IBM Watson to the Epic Sepsis Model
- **Cost:** Judge ground truth by the value of better decisions, not the cost of labels
- **Risk:** Experts disagree, proxies encode bias, synthetic data carries assumptions
- **Takeaway:** As models commoditize, proprietary, continuously validated ground truth is the differentiator

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Next week : Term Project Mid-Semester Presentations (Oct 14th)

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Recap from Last Class
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1. Alt data is always broken when it arrives. Cleaning is not a step before the real work - it is the real work.

2. Always beat the baseline first. Before calling any model an improvement, measure it against the reasonable alternative.

3. No single metric tells the whole story. RMSE answers: how large are my errors? MAE answers: what is my average miss? MAPE answers: how big is my miss relative to the actual? Hit ratio answers: am I predicting the right direction? Use all four.

4. The backtest is not the validation set. Walk-forward validation with quarterly retraining is the minimum standard. Look-ahead bias and survivorship bias both make results look better than they are.

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Out of class Activity Discussion

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Out of Class Activity : Step into Maya's shoes

You have been handed weekly sales data for a fast-casual sandwich chain - 3 years of history. You know the following:

- Sales spike every September (back-to-school) and dip every January
- The chain opened 40 new locations in Year 3, growing from 180 to 220 stores
- One month of foot traffic data is missing because the vendor had an outage
- A competitor closed 30 locations in the same markets in Year 2
- You have Google Trends data pulled across two separate API calls with different date windows



Let's discuss :



- What cleaning decisions will you make before touching any model, and why?
- Which baseline model would you start with - Naive, Seasonal Naive, or Holt's Trend - and what is your reason?
- What will RMSE alone fail to tell you about this dataset, and which other metric would you pair it with?
- Identify one period in this data where you expect any model to fail badly. What makes that period structurally different?
- What is the one number you must compute before reporting raw sales figures - and why does the store expansion make it non-optional?

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Investment Research with Alternative Data: From Models to Strategy

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You Have a Good Forecast. Now What?

What we have (Lec 6)

Daily Chipotle spend index from a card panel, normalized for store count

SARIMA beat Naïve, Seasonal Naïve and Holt on walk-forward validation

A decent read on comps weeks before the print

What sits between accuracy and returns

T **Target**

What exactly are we predicting — sales, or the thing the price reacts to?

R **Rule**

When the model says X, what do we do, on which day, for how long?

R **Risk**

How much, hedged against what, and after what costs?

Investors did not give us $10M for a low RMSE. They gave it to us for risk-adjusted returns.

Class Vote: Buy or Sell?

You know Chipotle's comparable sales before anyone else. Five real quarters.

Quarter	Reported	Comparable sales	Your call
Q2 2025	23 Jul 2025	−4.0%	BUY / SELL
Q3 2025	29 Oct 2025	+0.3%	BUY / SELL
Q4 2025	3 Feb 2026	−2.5%	BUY / SELL
Q1 2026	29 Apr 2026	+0.5%	BUY / SELL
Q2 2026	29 Jul 2026	+2.2%	BUY / SELL

Hands up for each row. Assume your card panel told you the comp number exactly, a week before the announcement.

The Reveal: The Stock Moved on the Outlook

Quarter	Comps	vs. the Street	Guidance	Stock reaction
Q2 2025	−4.0%	below	FY cut: growth → "about flat"	≈ −10% after hours
Q3 2025	+0.3%	revenue missed	FY cut: flat → low-single-digit decline	≈ −18% next day
Q4 2025	−2.5%	beat (est. −2.9%)	2026 "about flat" vs Street ≈ +1.8%	≈ −6% to −10%
Q1 2026	+0.5%	revenue beat	held "about flat"	≈ +3% after hours
Q2 2026	+2.2%	beat (est. +1.4%)	raised to low-single-digit growth	≈ +13%

Sign of comps called 4 of 5…

…and missed the biggest move. Q3 2025: comps positive, stock −18%.

Every cut or weak outlook: the stock fell

Q4 2025 beat on comps and still fell — on 2026 guidance.

Sources: Chipotle Q2 2025–Q2 2026 earnings releases (ir.chipotle.com); consensus and price reactions as reported by CNBC, Yahoo Finance/Zacks, Benzinga, Alpha Spread, Investing.com.

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Three Clocks — and Where the Money Is



Where returns concentrate

Around ~4 announcement days a year per stock. Daily trading on daily sales noise mostly pays spreads.

Satellite parking lots

Katona et al. (JFQA, 2025): 44 US retailers, 2011–17. Long-short on car counts earned ≈ 4–5% in the 3 days around earnings.

Chipotle, Q2–Q3 2025

Q2's CFO: sales turned positive in June. Q3's guide was about Oct–Dec. The data that mattered was the new quarter's.

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Choose the Target Before the Model

Each rung is closer to money and harder to measure



PITFALL

A naive strategy optimizes rung 1 and trades as if it were rung 5.

Nothing in RMSE tells you whether the Street already knew.

From Data to Alpha : Lululemon Sales Forecasting with Transaction Data

What's done (We covered this last class with Chipotle!) :



1. Cleaned raw transaction logs (missing values, outliers, merchant names)
2. Explored daily sales trends and volatility
3. Split data for back-testing (train/test) to avoid look-ahead bias
4. **SARIMA(1, 1, 2) × (1, 1, 2, 7)** delivered the lowest error and highest hit ratio.

The model winner:

SARIMA crushed the competition. **Best RMSE (8.59), highest hit ratio (81%),** followed actual sales patterns like magic. Our model testing phase is officially over.

But Here's the Reality Check:

Our investors didn't give us $10 million to publish academic papers or win modelling competitions. They want one thing: **superior returns on their money**.

The challenge: Bridge the gap between "our model predicts sales accurately" and "our investors get rich."

- So, we’ve built a model that predicts beautifully, but that’s only half the story.
- The real challenge now is transforming those predictions into impact, turning statistical accuracy into financial performance.
- In today's class, we’ll start exploring exactly how we make that leap from model output to money on the line.

The Trading Strategy

When SARIMA predicts big sales jump (>2%) → BUY Lululemon stock aggressively

- Logic: Strong sales → Excited investors → Stock price rises

When SARIMA predicts sales drop (>2%) → SELL or stay in cash

- Logic: Weak sales → Worried investors → Stock price falls

When prediction is small (<2%) → Do nothing

- Logic: Minor changes could be noise, not worth risking capital

Staying Fresh in Changing Markets

Why we can't get comfortable:

- Consumer habits change (pandemic shopping, inflation effects)
- Seasonal patterns shift (holiday timing, back-to-school trends)
- Economic conditions evolve (recession fears, interest rate changes)

The Ultimate Test - Backtesting Our $10M Strategy

What is Backtesting?

Test our exact strategy on real historical market data to see if we would have made or lost money.

The time travel test:

- Go back to March 2022 with $10 million and our SARIMA strategy.
- Live through the next 4 years making trades exactly as our rules dictate.
- No looking ahead, no second-guessing.

The final question: After 4 years, are our investors happy with their returns... or looking for new fund managers?

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Why This Test Matters So Much:

- **Investor trust**: People need proof before giving you millions to manage
- **Risk assessment**: How much could we lose in terrible market conditions?
- **Strategy validation**: Does our approach actually work in real markets?
- **Performance benchmark**: Can we beat simple index fund investing?

Our Testing Period: March 2022 - October 2024

Why These Dates Are Challenging:

March 2022: Russia invades Ukraine → Global market panic

Summer 2022: Inflation hits 40-year highs → Federal Reserve raises rates aggressively

Fall 2022: Recession fears dominate headlines → Growth stocks decline sharply

2023: AI revolution begins → Massive technology sector rotation

2024: Election uncertainty → Daily market volatility

Our Backtesting Process (Illustrative)

The setup:

- **Starting amount**: $10,000,000
- **Time period**: 16 quarters of trading
- **Model updates**: Retrain SARIMA every quarter
- **Position limits**: Maximum $500,000 per trade
- **Real conditions**: Include trading costs and market impact

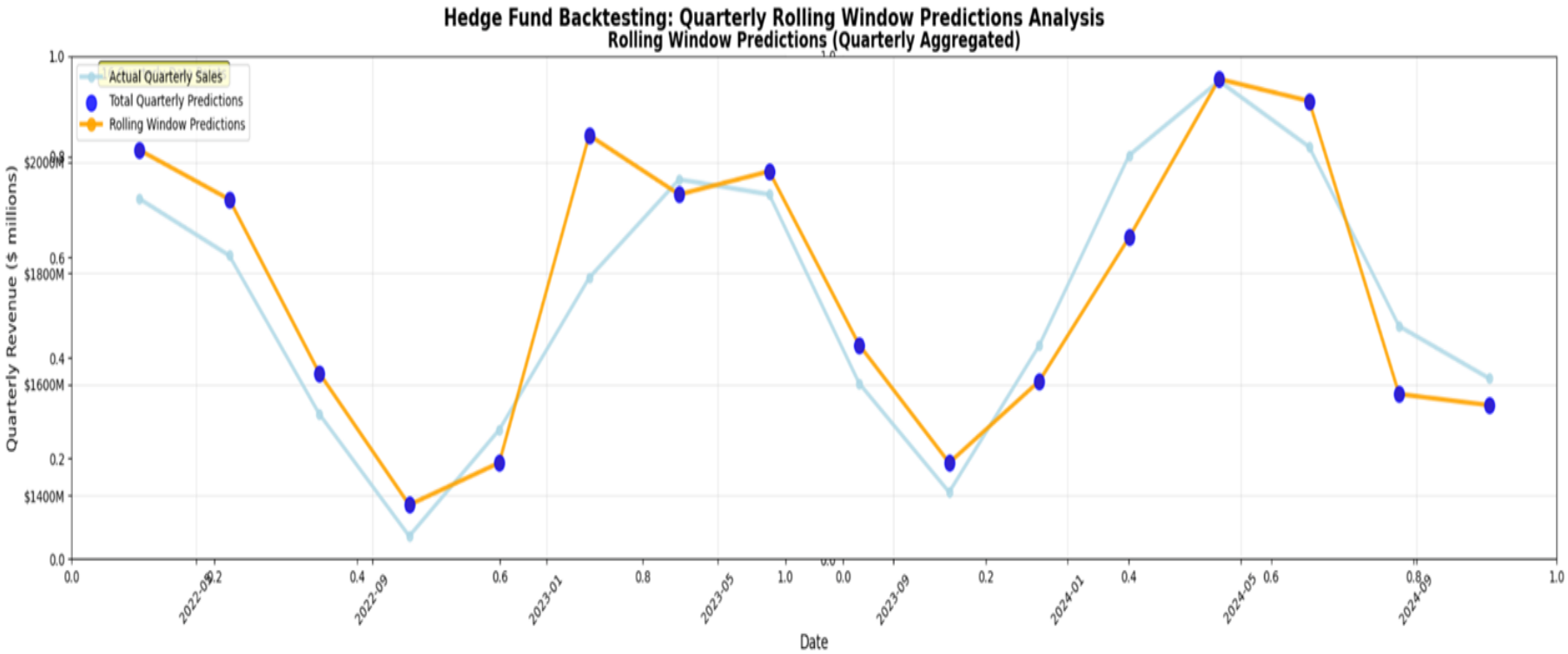
The process: Train model → Generate signals → Execute trades → Measure results → Repeat for 16 quarters

The moment of truth: Did we grow our investors' $10 million or lose it?

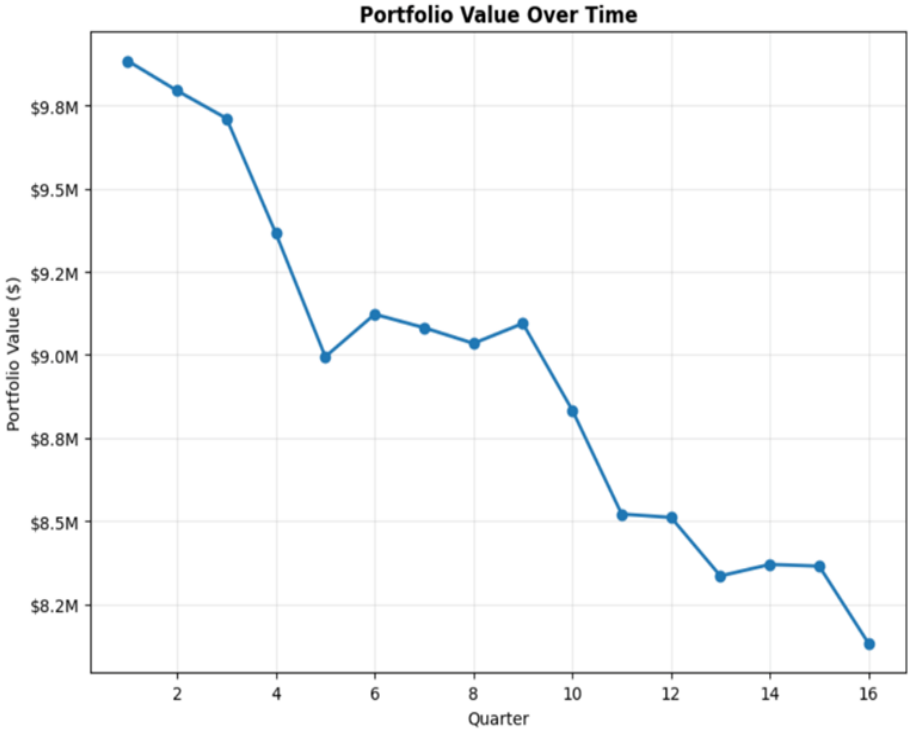
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QUARTERLY RESULTS SUMMARY
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```

End_Date	Model	RMSE	MAPE	Hit_Ratio	Quarter_Return	Portfolio_Value	Cumulative_Return
2022-03-22	SARIMA((2, 1, 2), (1, 1, 1, 7))	9.542209	8.600127	45.0	-1.162587	9.883741e+06	-1.162587
2022-05-24	SARIMA((2, 1, 2), (1, 1, 1, 7))	8.945431	9.710095	45.0	-0.906119	9.794183e+06	-2.058171
2022-07-26	SARIMA((2, 1, 2), (1, 1, 1, 7))	5.696621	6.242943	60.0	-0.850039	9.710929e+06	-2.890714
2022-09-27	SARIMA((2, 1, 2), (1, 1, 1, 7))	7.796883	10.477431	55.0	-3.559520	9.365266e+06	-6.347339
2022-11-29	SARIMA((2, 1, 2), (1, 1, 1, 7))	6.313672	6.997152	60.0	-3.957099	8.994673e+06	-10.053267
2023-01-31	SARIMA((2, 1, 2), (1, 1, 1, 7))	13.451179	15.008993	70.0	1.419207	9.122326e+06	-8.776737
2023-04-04	SARIMA((2, 1, 2), (1, 1, 1, 7))	5.958961	5.419725	60.0	-0.443617	9.081858e+06	-9.181419
2023-06-06	SARIMA((2, 1, 2), (1, 1, 1, 7))	7.993247	7.264031	65.0	-0.518979	9.034725e+06	-9.652748
2023-08-08	SARIMA((2, 1, 2), (1, 1, 1, 7))	7.211393	7.402044	55.0	0.662919	9.094618e+06	-9.053819
2023-10-10	SARIMA((2, 1, 2), (1, 1, 1, 7))	7.203162	9.111787	50.0	-2.862659	8.834270e+06	-11.657299
2023-12-12	SARIMA((2, 1, 2), (1, 1, 1, 7))	8.180832	8.365444	65.0	-3.537843	8.521728e+06	-14.782725
2024-02-13	SARIMA((2, 1, 2), (1, 1, 1, 7))	8.869568	7.747271	70.0	-0.118046	8.511668e+06	-14.883320
2024-04-16	SARIMA((2, 1, 2), (1, 1, 1, 7))	5.418001	4.571376	75.0	-2.066580	8.335768e+06	-16.642325
2024-06-18	SARIMA((2, 1, 2), (1, 1, 1, 7))	6.533530	5.756668	75.0	0.414238	8.370297e+06	-16.297026
2024-08-20	SARIMA((2, 1, 2), (1, 1, 1, 7))	9.363263	8.693801	60.0	-0.061578	8.365143e+06	-16.348568
2024-10-22	SARIMA((2, 1, 2), (1, 1, 1, 7))	5.640095	5.779039	75.0	-2.777502	8.132801e+06	-18.671988

```
============================================================
HEDGE FUND PERFORMANCE METRICS
============================================================
Total Return %: -18.67
Annual Return %: -15.02
Volatility %: 4.11
Sharpe Ratio: -4.38
Max Drawdown %: 18.67
Hit Rate %: 18.75
Final Portfolio Value: $8,132,801
Total Quarters: 16
```



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The Shocking Results - Our $10M Reality Check

What Actually Happened After 16 Quarters

The Final Numbers :

```
Started with:     $10,000,000
Ended with:       $8,132,801
We Lost:          $1,867,199
Total Return:     -18.67%
```

The brutal truth: We turned our investors' $10 million into $8.1 million. Nearly $2 million simply vanished.

The Investors' Phone Call:

You said SARIMA was the best model. How did you lose our retirement money?

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The Quarter-by-Quarter Agony

Quarters 1-5: The Steady Bleed

- Started at $10M → Dropped to $8.9M
- Small losses every quarter (-1% to -4%)
- Investor mood: Optimistic → Concerned → Worried

Quarter 6: False Hope

- **+1.42%** *(our only decent win)*
- Everyone celebrated: "Finally it works!"

Quarters 7-16: The Grinding Reality

- Back to consistent losses
- Two tiny wins: +0.66%, +0.23% *(meaningless)*
- Final decline: $8.9M → $8.1M

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When we share these results with professional investors, they always ask the same question: *"What's your Sharpe ratio?"*

Understanding Sharpe Ratio

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The basic question: *"How much return do you earn for each unit of risk you take?"*

Real-world analogy:

- **Job A**: $50,000 salary, guaranteed, no stress
- **Job B**: $60,000 expected, but 50% chance of getting fired, high stress
- **Question**: Is the extra $10,000 worth the risk and sleepless nights?

Sharpe Ratio answers this for investments.

Our Step-by-Step Calculation:

Step 1: Gather the inputs from our backtesting

- **Portfolio Return**: -15.02% per year *(we lost money)*
- **Risk-Free Return**: +3% per year *(government bonds - safe alternative)*
- **Portfolio Volatility**: 4.11% *(how much our returns bounced around daily)*

Step 2: Calculate the numerator (excess return)

```
Excess Return = Portfolio Return - Risk-Free Return
Excess Return = -15.02% - 3% = -18.02%
```

Translation: We performed 18.02% worse than the safe alternative

Step 3: Apply the complete formula

```
Sharpe Ratio = Excess Return ÷ Portfolio Volatility
Sharpe Ratio = -18.02% ÷ 4.11% = -4.38
```

Step 4: Interpret the result

- **Negative Sharpe**: We lost money while taking risk
- **- 4.38 magnitude**: We lost A LOT of money while taking risk

The Professional Grading Scale:

```
Sharpe > 2.0:   A+ (Warren Buffett territory)
Sharpe 1.5-2.0: A  (Top hedge funds)
Sharpe 1.0-1.5: B  (Solid professional level)
Sharpe 0.5-1.0: C  (Barely acceptable)
Sharpe 0.0-0.5: D  (Taking risk for no reward)
Sharpe < 0.0:   F  (Losing money while stressed)
```

What Our - 4.38 Actually Means:

"For every 1% of risk we took, we lost 4.38% compared to safe investments. We gave investors stress AND losses."

How Did SARIMA (A Perfect Model) Fail?

The discovery from our Backtesting:

There was **almost zero correlation** between model accuracy and trading profits.

Examples from our actual results:

- **Quarter 13**: RMSE 5.42 *(nearly perfect sales predictions)* : Lost 2.07%
- **Quarter 6**: RMSE 13.45 *(terrible predictions)* : Made 1.42% *(our best quarter!)*
- **Quarter 16**: RMSE 5.64 *(excellent predictions)* : Lost 2.78%

The realization

Problem we mastered: *"How to predict Lululemon daily sales accurately"*

Result: SARIMA was genuinely excellent at this

Problem we needed to solve: *"How to predict Lululemon stock price movements"*

Result: We failed catastrophically at this

The Critical Insight That Changes Everything:

Predicting sales ≠ Predicting stock prices

They're as different as predicting tomorrow's weather vs predicting what clothes everyone will choose to wear.

This discovery leads us to the next crucial question: What actually does move stock prices?

What Actually Moves Stock Prices - The Reality

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What we focused on:

- Daily Lululemon sales accuracy
- SARIMA prediction precision
- Historical patterns and seasonality
- *"Good sales numbers = Happy investors = Higher stock price!"*

What Actually Dominated Markets During 2022-2024

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2022: The Perfect Storm

- **March**: Russia invades Ukraine → Global panic, ALL stocks crash regardless of sales
- **Summer**: Inflation hits 40-year highs → Nobody cares about retail fundamentals
- **Fall**: Recession fears dominate → "Good sales won't matter if economy collapses"

2023: The AI Revolution

- **ChatGPT launches** → Investors only want AI stocks, retail becomes "old economy"
- **Sector rotation** → Money flows OUT of consumer stocks INTO technology
- **Our situation**: Perfect Lululemon predictions ignored by AI-obsessed market

2024: Political and Economic Chaos

- **Election uncertainty** → Daily market swings based on polls, not earnings
- **Federal Reserve confusion** → Interest rate fears trump all company data

What We Actually Learned

During our **backtesting** period, stock prices seemed to be driven more by:

- **Market-wide events** *(wars, inflation, AI hype, elections)*
- **Than by individual company fundamentals** *(our sales predictions)*

Evidence from our results:

- **Best prediction quarters** often had **worst trading results**
- **Worst prediction quarters** sometimes had **best trading results**
- **Almost no correlation** between SARIMA accuracy and profits

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If even perfect predictions don't guarantee profits, is there a better way to use alternative data?

Answer: Yes! Our approach was naive, but professional hedge funds have developed sophisticated methods that actually do make alternative data profitable.

How Professional Hedge Funds Actually Do It

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Think of It Like School vs Professional Sports

Our Approach Was Like High School Basketball:

- **One star player:** We only used SARIMA (like having one good player)
- **Simple plays**: "If ball is here, do this" (like our "if sales up, buy stock")
- **Same strategy always**: Ran same plays whether winning or losing
- **Fixed effort**: Always played at same intensity regardless of situation

Professional Hedge Funds Are Like NBA Teams:

- **Multiple star players**: Different models for different situations
- **Complex playbook**: Hundreds of different strategies depending on conditions
- **Adaptive coaching**: Change strategy if losing, different plays for different opponents
- **Variable intensity**: Go all-out when opportunity is great, play defense when necessary

Let's Learn Each Professional Technique Step by Step

Technique 1: Use Multiple Models (Like Having a Full Team)

What we did wrong:

- Only used SARIMA for everything
- When SARIMA had a bad quarter, our entire strategy failed
- Like a basketball team with only one player

What professionals do:

- Use 3-4 different models: RandomForest, SARIMA, Ensemble Modelling
- Each quarter, test all models and pick the best performer
- If one model fails, others can still succeed

Why this works: Like having multiple players - if your star player is having an off day, your backup can still win the game.

Technique 2: Smart Signal Types (Not Just Buy/Sell/Hold)

What we did:

- **Simple binary signals**: Sales up >2% = Buy, Sales down >2% = Sell, Otherwise = Hold
- **Only considered sales predictions**: Ignored everything else happening in markets
- **Same signal logic always**: Whether markets were calm or crashing, we used identical decision rules

What professionals do differently:

- **Multiple signal types**: Instead of just buy/sell/hold, they might have different strength levels
- **Multi-factor signals**: Combine sales predictions with market momentum, volatility, and sentiment
- **Context-aware signals:** Different signal logic for different market conditions

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Our Approach:

- SARIMA predicts sales up >2% → Buy signal (regardless of anything else)

Professional Approach:

- SARIMA predicts sales up >2% + stock momentum is positive + market volatility is low → Strong buy signal
- SARIMA predicts sales up >2% + stock momentum is negative + market volatility is high → Weak buy signal or hold
- SARIMA predicts sales up >2% + market is in crisis mode → Hold (wait for better conditions)

Technique 3: More Sophisticated Position Sizing

What we actually did:

- **Fixed maximum limit:** Never exceeded $500K (5% of fund) no matter what
- **Single-factor scaling:** Position size only considered SARIMA confidence (lower RMSE = larger position)
- **Ignored market context:** Used same position sizing approach whether markets were calm or crashing

What professionals do differently:

- **Dynamic maximum limits:** The maximum they're willing to risk changes with market conditions
- **Multi-factor scaling**: Consider model confidence AND market volatility AND overall conditions
- **Market-aware sizing**: Dramatically reduce position sizes during crisis periods

Technique 4: Change Strategy Based on Market Conditions

What we did wrong:

- Used same approach whether markets were going up, down, or crashing
- Like wearing summer clothes in winter because "clothes are clothes"

What professionals do:

- **Bull Market** (everything going up): Be aggressive, fundamentals matter
- **Bear Market** (everything going down): Be defensive, protect capital
- **Crisis** (panic everywhere): Go to cash, wait for stability
- **Recovery** (coming out of crisis): Gradually increase risk

The Big Picture - Why Professionals Succeed

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They Think in Layers:

1. **Layer 1**: Is our model accurate this quarter?
2. **Layer 2**: What are market conditions like?
3. **Layer 3**: How confident should we be?
4. **Layer 4**: What's the right position size?
5. **Layer 5**: Should we trade at all right now?

What happens when we apply ALL these professional techniques to our same Lululemon data that lost us $2 million?

The Enhanced Strategy Framework:

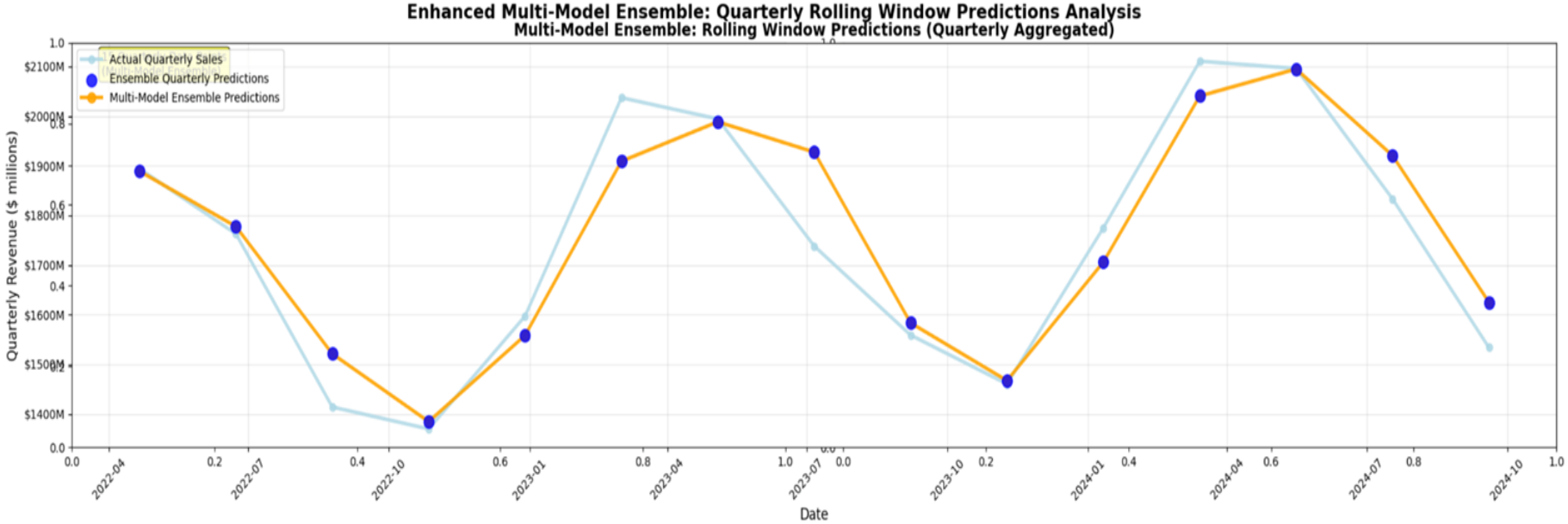
Multi-model ensemble with dynamic risk management and enhanced signal processing.

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================================================================================
MULTI-MODEL ENSEMBLE: QUARTERLY RESULTS SUMMARY
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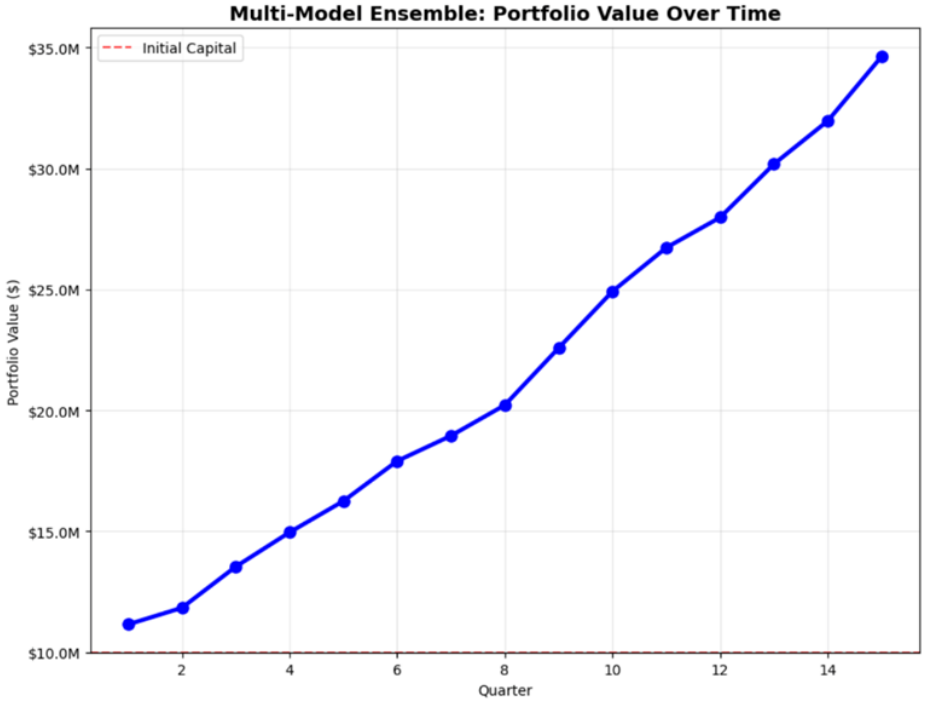
End_Date	Model	RMSE	MAPE	Directional_Accuracy	Avg_Position_Size	Quarter_Return	Portfolio_Value
2022-04-21	Ensemble[RandomForest, PolyLinear, ARIMA]	2.47	2.24	95.00	9.03	11.44	11144304.16
2022-06-23	Ensemble[RandomForest, PolyLinear, ARIMA]	3.93	3.95	90.00	5.63	6.15	11830126.67
2022-08-25	Ensemble[RandomForest, PolyLinear, ARIMA]	6.52	8.91	95.00	6.34	14.38	13531669.70
2022-10-27	Ensemble[RandomForest, PolyLinear, ARIMA]	3.16	3.90	100.00	5.07	10.50	14952118.82
2022-12-29	Ensemble[RandomForest, PolyLinear, ARIMA]	2.86	2.97	100.00	5.46	8.64	16244504.39
2023-03-02	Ensemble[RandomForest, PolyLinear, ARIMA]	8.35	6.33	85.00	6.13	10.20	17900649.82
2023-05-04	Ensemble[RandomForest, PolyLinear, ARIMA]	2.56	1.97	95.00	4.99	5.84	18946482.26
2023-07-06	Ensemble[RandomForest, PolyLinear, ARIMA]	9.91	11.28	90.00	5.45	6.69	20214936.42
2023-09-07	Ensemble[RandomForest, PolyLinear, ARIMA]	3.03	3.23	90.00	6.14	11.68	22576253.22
2023-11-09	Ensemble[RandomForest, PolyLinear, ARIMA]	2.41	2.83	95.00	4.93	10.37	24916780.74
2024-01-11	Ensemble[RandomForest, PolyLinear, ARIMA]	4.53	4.25	90.00	5.59	7.25	26723530.09
2024-03-14	Ensemble[RandomForest, PolyLinear, ARIMA]	4.70	3.76	95.00	5.66	4.68	27973629.95
2024-05-16	Ensemble[RandomForest, PolyLinear, ARIMA]	2.08	1.69	95.00	7.33	7.89	30181475.86
2024-07-18	Ensemble[RandomForest, PolyLinear, ARIMA]	5.05	5.18	85.00	5.51	5.91	31966570.62
2024-09-19	Ensemble[RandomForest, PolyLinear, ARIMA]	4.88	6.30	85.00	5.90	8.33	34629168.34

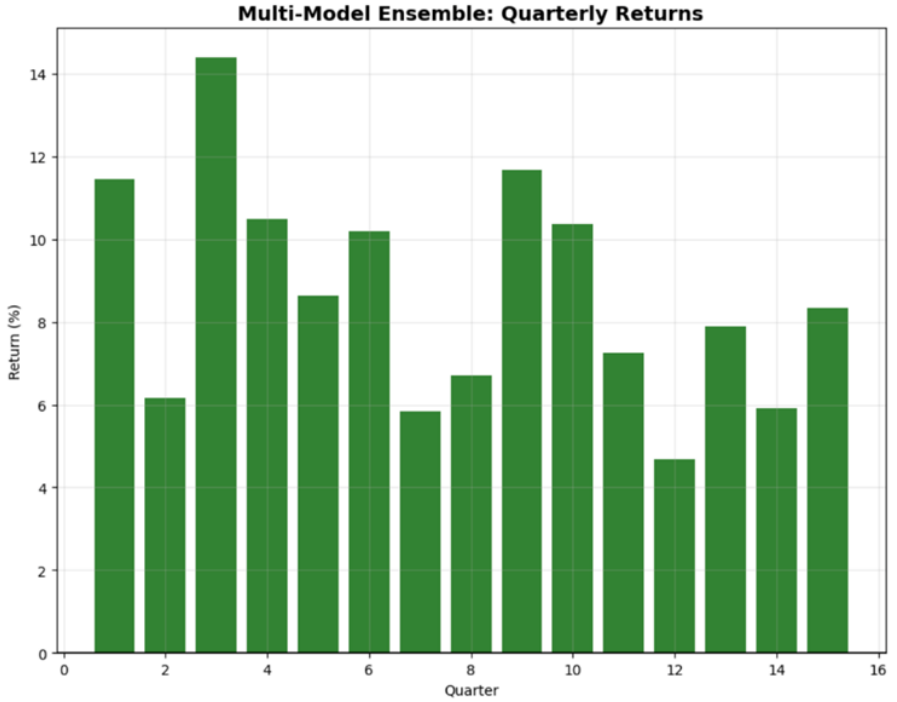
```
================================================================================
MULTI-MODEL ENSEMBLE: HEDGE FUND PERFORMANCE METRICS
================================================================================
Total Return %: 246.29
Annual Return %: 183.88
Volatility %: 9.13
Sharpe Ratio: 11.74
Max Drawdown %: 0.31
Win Rate %: 100.0
Avg Quarterly Return %: 8.66
Final Portfolio Value: $34,629,168
Total Quarters: 15
Profit Loss: $24,629,168
```

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Key Performance Metrics

- **Sharpe Ratio: 11.74** (exceptional risk-adjusted performance)
- **Maximum Drawdown: 0.31%** (superior risk control)
- **Win Rate: 100%** (15 profitable quarters out of 15)
- **Annual Return: 183.88%** (outstanding alpha generation)
- **Average Quarterly Return: 8.66%**
- **Volatility: 9.13%** (controlled risk level)

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Enhanced Strategy Components

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Model Ensemble Approach

Multiple Models Combined Each Quarter:

- **RandomForest**: Primary machine learning algorithm
- **Polynomial Linear Regression**: Enhanced regression with polynomial features
- **ARIMA**: Time series fallback option
- **Combination Method**: Weighted ensemble based on performance scores (not selection)

Actual Results: All three models used together with dynamic weights across 15 quarters

Performance Range: RMSE values from 2.08 to 9.91 across quarters

Directional Accuracy: 85-100% in all quarters

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Enhanced Feature Engineering

Technical Indicators Created:

- **Moving Averages** (5-day, 10-day, 20-day averages): *Shows if sales are trending up or down over different time periods - helps identify patterns*
- **Momentum Features** (3-day and 7-day price change percentages): *Measures how fast sales are changing - tells us if momentum is building*
- **Volatility Measures** (5-day and 10-day rolling standard deviations): *Shows how much sales bounce around - helps us understand risk level*

Enhanced Feature Engineering

- **Position Features** (Price relative to moving averages): *Tells us if current sales are above or below recent averages - indicates strength*
- **Time Features** (Day of week, month, quarter): *Captures seasonal patterns - weekends vs weekdays, holiday seasons*
- **Lag Features** (1, 2, 3, 5, and 7-day historical values): *Uses past sales to predict future - because yesterday's sales influence today's*

Why use all these? Simple sales numbers alone don't tell the full story. These features help the model understand *context*- is this a normal day or unusual? Is momentum building or fading?

Advanced Signal Processing

Multi-Factor Signal Types:

- **STRONG_LONG** (Price change >8% + positive momentum >5%): *Big predicted jump AND momentum building = Very confident buy*
- **LONG** (Price change >6%): *Moderate predicted increase = Somewhat confident buy*
- **STRONG_SHORT** (Price change <-8% + negative momentum <-5%): *Big predicted drop AND momentum falling = Very confident sell*
- **SHORT** (Price change <-6%): *Moderate predicted decrease = Somewhat confident sell*
- **HOLD**: *Small changes or conflicting signals = Don't trade, wait for clarity*

Why different signal strengths? Like weather forecasting - there's a difference between "might rain" and "hurricane coming." We bet more when very confident, less when uncertain.

Context Filters Applied

Momentum threshold 5%: *Don't make big bets unless momentum confirms the prediction*

Volatility adjustment with 8x penalty: *Heavily reduce bet sizes when markets are chaotic*

Multi-factor confirmation: *Require multiple indicators to agree before trading*

Why use filters? Like having safety checks before making important decisions. We don't want to bet big just because one indicator looks good, we need multiple confirmations that conditions are right.

Dynamic Position Sizing

Goal: Adjust bet size based on how confident you are and how risky conditions are.

The Three-Factor Decision Process

Factor 1: Signal Strength (How Strong is Our Prediction?)

STRONG_LONG/SHORT (4% maximum):

- Our model predicts Lululemon stock will move >8%
- AND momentum indicators confirm the direction
- AND volatility is under control
- Think: "We're very confident = willing to risk more"
- Example: Model says +9% move, momentum is positive, low volatility

LONG/SHORT (2.4 % maximum):

- Model predicts moderate move > 6% but < 8%
- OR momentum doesn't fully confirm
- Think: "We're somewhat confident = risk less"
- Example: Model says +7% move but momentum is mixed

HOLD (0%):

- Model prediction is <6%
- OR conflicting signals from different indicators
- Think: "We're not sure = don't risk anything"

Factor 2: Model Confidence Score

How it works:

- Multi-model ensemble gives confidence score from 0.0 to 1.0
- 0.9+: Very confident in prediction
- 0.7-0.8: Moderately confident
- <0.6: Low confidence

Real Example:

- Signal: STRONG_LONG (4% base)
- Model confidence: 0.8 (pretty sure)
- So far: 4% × 0.8 = 3.2%

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Factor 3: Volatility Adjustment

Market Volatility Check:

- **Low volatility** (calm markets): Adjustment = 1.0 (no reduction)
- **Medium volatility:** Adjustment = 0.3-0.5 (moderate reduction)
- **High volatility (chaotic markets):** Adjustment = 0.1-0.2 (massive reduction)

Continuing Example:

- Current position: 3.2%
- Market volatility is medium: 0.4 adjustment
- Final position: 3.2% × 0.4 = **1.3%**

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Results From Our New Strategy

Quarterly Position Sizes:

- **Q1 (April 2022):** 9.03% - Great conditions, high confidence
- **Q7 (April 2023):** 4.99% - Moderate conditions
- **Q12 (Feb 2024):** 5.66% - High volatility period, system reduced risk
- **Q15 (Aug 2024):** 5.90% - Good but not perfect conditions

Why Never Hit 4% Maximum?:

Real markets are never perfect. The system found reasons to be cautious:

- Model confidence wasn't always 100%
- Volatility was often elevated
- Multiple safety checks prevented overconfidence

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The Alpha Boost - Your Information Edge

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What is Alpha Boost?

Alpha Boost = The extra profit from having better information than everyone else

In our Lululemon strategy: 2 basis points (0.02%) daily advantage from superior sales data

How It Works?

Trading Lululemon Stock on the Same Day:

- **Other traders**: Use public news, charts, guesswork → Make 0.5% profit
- **You with sales data**: Same trade BUT with real sales numbers → Make 0.52% profit
- **Your edge**: +0.02% extra every day you trade

Why Do You Get This Edge?

- You have actual Lululemon sales data before it's public
- Other traders are making educated guesses
- Better information = better timing and decisions

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The Compounding Power

Over 15 Quarters:

- **Total trading days**: ~975 days
- **Days you were positioned**: ~70% = ~680 days
- **Extra return from data edge**: 680 × 0.02% = **~13.6% bonus**

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Professional Implementation Insights

Key Technical Advantages

Feature Engineering Impact: The 15+ engineered features (moving averages, momentum, volatility, lags) provided the **multi-model ensemble** with rich information beyond basic sales data, enabling superior pattern recognition across different algorithms.

Ensemble Model Benefits: Combining multiple algorithms (RandomForest, Polynomial Linear, ARIMA) with dynamic performance-weighted averaging ensured robust predictions and reduced single-model risk, though the ensemble's collective strength validated the effectiveness of this diversified approach for this dataset.

Key Technical Advantages

Signal Processing Enhancement: Multi-factor signal generation (price prediction + momentum + volatility) reduced false signals and improved trade timing compared to simple threshold approaches.

Dynamic Risk Adaptation: Position sizing that considered model confidence, signal strength, and portfolio volatility prevented large losses while enabling significant gains during favorable periods.

Alternative Data Value Realization

Information Edge Monetization:

- **2 basis points daily alpha** when positioned
- **Systematic capture** of alternative data advantage through professional execution
- **Consistent application** across all 15 quarters regardless of market conditions

Bottom Line: The alpha boost proves that alternative data has real, measurable value when executed correctly. This is the difference between having data and professionally monetizing it.

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The Complete Value Chain

1. **Get superior data** (Lululemon sales)
2. **Build better models** (RandomForest + Polynomial Linear + ARIMA)
3. **Execute professionally** (dynamic position sizing)
4. **Capture information edge** (alpha boost)

Professional Framework Validation: The strategy successfully transformed the same sales data from a losing proposition into exceptional returns through sophisticated implementation, demonstrating that alternative data value depends heavily on execution quality.

What did we Learn

Foundations

- **Hedge Funds Basics** – Introduced what hedge funds are and how they generate alpha using active, risk-managed strategies.
- **Alternative Data Edge** – Explored how sources like satellites, credit cards, and web traffic give investors a unique early advantage.
- **Data Cleaning** – Prepared raw transaction logs by normalizing names, fixing missing values, and filtering for relevant insights.
- **Exploring Trends** – Studied Lululemon's daily sales to uncover patterns, spikes, and meaningful signals.

Modelling Approaches

- **Baseline Forecasts** – Naïve and Seasonal Naïve models gave us starting benchmarks.
- **Advanced Models** – Holt's Trend, ARIMA, and SARIMA helped capture both short-term dynamics and seasonal cycles.

Strategy & Testing

- **Backtesting Strategies** – Tested strategies with simulated capital to connect predictions to trading outcomes.
- **Importance** – Saw the importance of aligning predictive accuracy with real market dynamics for effective results.

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Hedge Fund Playbook

- **Techniques** – Learned how hedge funds use multi-model ensembles, layered signals, and adaptive risk strategies.
- **Execution Enhancements** – By applying feature engineering, multi-factor signals, and context filters, strategies became more robust.
- **Complete Value Chain** – Superior data, stronger models, disciplined execution, and capturing measurable alpha.

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Thought Leadership Article : Due Nov 5th

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Reflection Assignment: Thought Leadership Piece

Each student will prepare an individual thought leadership article (600–900 words) that explores a recent advancement or research development in the field of alternative data.

Goal:
Translate cutting-edge trends, research, or innovations into a practical, accessible piece that business professionals, investors, and product peers can learn from and apply.

Format:
Students may choose to submit on Canvas or, if they wish, share publicly (e.g., LinkedIn, Medium). Posting publicly is optional, not required.

Guidelines:

- Audience: Business professionals and decision-makers interested in alternative data
- Tone: Practical, engaging, and forward-looking
- Content Focus:
- Select a recent advancement, innovation, or research study in alternative data (e.g., investment research applications, synthetic data applications, ESG datasets, multimodal fusion, regulatory shifts)
- Explain the problem it addresses and why it matters
- Show how it can be applied in practice (with an example or scenario)
- End with clear, actionable insights for practitioners

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Next week : Term Project Mid-Semester Presentations (Oct 14th)

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Overview

- Purpose of the Mid-Semester Presentation (15% of grade):
  - Present your team's progress since the Idea Pitch, highlighting what's been accomplished, what challenges have emerged, and what comes next.
- Total Time: 10 Minutes (HARD STOP)
  - You **must finish within 10 minutes total.**

Topics to include (Suggested outline but topics marked * are mandatory)

- **Project Overview***
  - Briefly restate the goals and objectives of your project from the introductory presentation.
  - Remind the audience of the problem you are solving and the intended impact.
- **Progress Overview***
  - Summarize the progress made since the start of the semester.
  - Highlight **key milestones achieved** and any **major challenges encountered.**
- **Customer Discovery**
  - Describe the interviews that you conducted with potential customers (Could be fellow Duke students!) to understand if there is a genuine need for your product and if the users requested additional features or did not agree with a few features.

- **Data Collection & Preparation***
  - Describe how data was collected, cleaned, and prepared.
  - Mention any issues such as missing data, bias, or limited availability, and how your team addressed them.
- **Model Development***
  - Outline the AI/ML models, algorithms, or analytical methods used so far.
  - Make sure to include the details of your BASELINE model.
  - Discuss **techniques, methodologies, and any preliminary results** obtained.
- **Evaluation Metrics***
  - Explain the **metrics** used to evaluate performance (e.g., accuracy, F1-score, RMSE, AUC).

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- **How you used the Plause peer Feedback to improve your idea/presentation. (While this is not required, it is highly encouraged!)**
- **Next Steps***
  - Outline upcoming tasks: experiments, optimizations, feature improvements, or deployment goals.
- **Challenges & Risks***
  - Identify current or potential obstacles that could affect progress.
  - Discuss **risk mitigation plans** or alternate strategies.

Tips for a Strong Presentation

- Keep slides clear, concise, and visually appealing.
- Use data visuals, screenshots, or short demos instead of heavy text.
- Be transparent about challenges, focus on how you're solving them.
- Transition smoothly between speakers.
- End with a clear "what's next" slide to signal momentum.
- Rehearse to ensure you stay within the 10-minute hard stop.

Logistics & Submission

- Deadline: 1:30 PM, Wednesday, October 14th (just before class).
- A link will be posted on Canvas to upload your presentation.
- Format: Only PDF (Please convert your ppts into pdfs for easy access)
- Review these guidelines carefully before submitting your final deck.

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Thank You & Questions

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Back up